

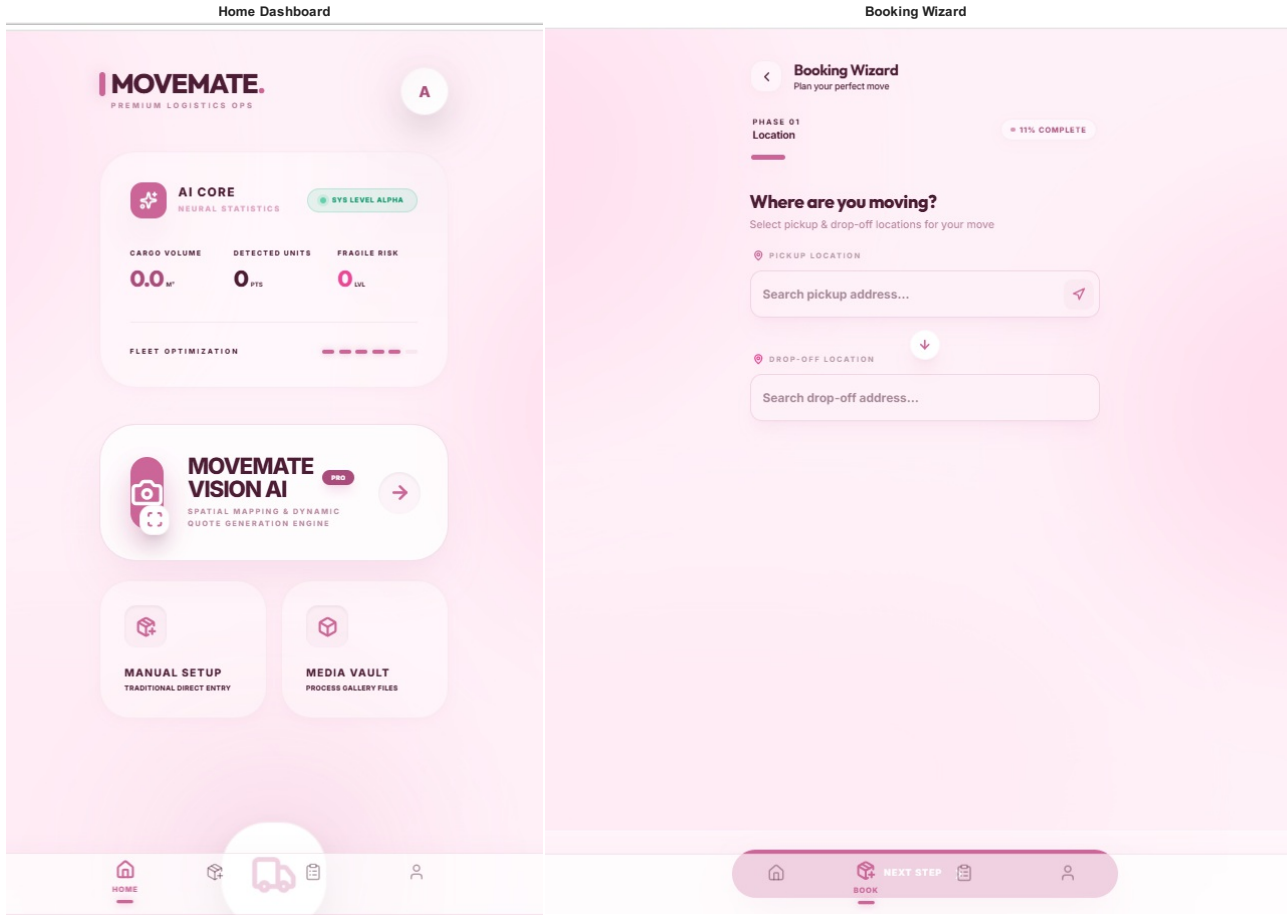
MoveMate Pro

A premium, AI-powered Progressive Web App (PWA) for booking movers and packers services.

MoveMate Pro reimagines the moving experience with a sleek mobile-first interface, intelligent inventory scanning, real-time pricing, and seamless booking — all wrapped in a beautiful, tactile design system.

Built with Next.js 16, Firebase, Google Gemini AI, and Google Maps.

Screenshots



What Does This App Do?

MoveMate Pro allows users to **book a complete moving service** from start to finish through a guided 9-step wizard:

1. **Choose Move Type** — Select between Home, Office, Single-Item, or Vehicle transport.
2. **Set Locations** — Pick pickup & drop-off points on an interactive Google Map with autocomplete.
3. **Property Details** — Specify floor levels, elevator access, parking distance, and narrow-street conditions.
4. **Build Inventory** — Add items manually by category or **scan a room with your camera** using AI to auto-detect furniture.
5. **Select Services** — Toggle professional packing, unpacking, insurance, and express delivery.
6. **Schedule** — Pick a date and time slot (AM / PM / Evening) or mark as urgent.
7. **Apply Discounts** — Enter promo codes for instant savings (e.g., MOVE10, FIRST20).
8. **Review Pricing** — See a transparent, itemized price breakdown in real-time.
9. **Confirm & Book** — Execute the booking, stored securely in Firebase.

After booking, users can:

- **Track moves live** with an animated truck on a visual map timeline.
- **View booking history** from their dashboard.
- **Manage their profile** — saved addresses, settings, and account info.

Key Features

AI-Powered Inventory Scanning

Point your camera at a room and let **Google Gemini 1.5 Flash** identify all furniture, electronics, and fragile items automatically. The AI returns structured data including item names, quantities, fragile status, and estimated volume — all fed directly into the pricing engine.

Smart Pricing Engine

A sophisticated pricing calculator that factors in:

- Base fare + distance rate (€3.50/km)
- Per-item handling costs (20+ item types with individual rates)
- Fragile item surcharge (+20%)
- Floor charges (with/without elevator)
- Service add-ons (packing, unpacking, insurance, express)

- Tax calculation (8%)
- Coupon/discount support

Real-Time Location with Google Maps

- Interactive map with draggable markers
- Google Places Autocomplete for address search
- Auto-calculated distance & estimated travel duration
- Visual route between pickup and drop-off

Live Order Tracking

Simulated real-time tracking page with:

- Animated truck moving along the route
- Status timeline (Assigned → On the Way → Arrived → Completed)
- Driver info card with call button

Premium UI/UX

- Light-themed, brand-aligned design system (primary: #cc6699)
- Glassmorphism cards with backdrop blur
- Reactive mesh background with mouse-tracking parallax
- Micro-animations: bounce-in, slide-up, pulse-glow
- Tactile button feedback
- Fully responsive, mobile-first layout

Project Architecture

```
movemate-pro/
├── src/
│   ├── app/
│   │   ├── page.tsx           # Home/Dashboard
│   │   ├── layout.tsx        # Root layout with providers
│   │   ├── globals.css       # Design system & CSS variables
│   │   ├── login/page.tsx    # Firebase Auth login
│   │   ├── profile/page.tsx  # User profile & saved addresses
│   │   ├── bookings/page.tsx # Booking history
│   │   ├── tracking/[id]/page.tsx # Live tracking view
│   │   └── book/
│   │       ├── page.tsx      # Booking wizard controller
│   │       └── steps/
│   │           ├── MoveTypeStep.tsx
│   │           ├── LocationStep.tsx
│   │           ├── PropertyStep.tsx
│   │           ├── InventoryStep.tsx
│   │           ├── ServicesStep.tsx
│   │           ├── ScheduleStep.tsx
│   │           ├── DiscountStep.tsx
│   │           ├── PricingStep.tsx
│   │           └── ConfirmStep.tsx
│   ├── actions/
│   │   └── ai-scan.ts        # Gemini AI server action
│   ├── components/
│   │   ├── MeshBackground.tsx # Animated parallax background
│   │   ├── CameraScan.tsx    # Camera capture + AI analysis
│   │   └── BottomNav.tsx     # Mobile bottom navigation
│   └── lib/
│       ├── firebase.ts       # Firebase app initialization
│       ├── auth-context.tsx  # Auth provider (React Context)
│       ├── booking-context.tsx # Booking state (React Context)
│       ├── firestore.ts      # Firestore CRUD operations
│       └── pricing.ts        # Pricing engine & coupon validation
├── firebase.json             # Firebase Hosting & Firestore config
├── firestore.rules           # Security rules
├── firestore.indexes.json    # Composite indexes
└── package.json
```

Current Tech Stack

Layer	Technology	Purpose
Framework	Next.js 16 (https://nextjs.org/) (App Router + Turbopack)	Server/client rendering, routing, server actions
Language	TypeScript 5 (https://www.typescriptlang.org/)	Type-safe development
UI Library	React 19 (https://react.dev/)	Component-based UI
Styling	Tailwind CSS v4 (https://tailwindcss.com/)	Utility-first CSS with custom design tokens
Icons	Lucide React (https://lucide.dev/)	Beautiful, consistent icon set
AI / ML	Google Gemini 1.5 Flash (https://ai.google.dev/) via @google/generative-ai	Vision-based inventory scanning
Maps	Google Maps JavaScript API (https://developers.google.com/maps) via @react-google-maps/api	Interactive maps, autocomplete, geocoding
Auth	Firebase Authentication (https://firebase.google.com/products/auth)	Google Sign-In, session management
Database	Cloud Firestore (https://firebase.google.com/products/firestore)	Real-time NoSQL document database
Hosting	Firebase Hosting (https://firebase.google.com/products/hosting)	CDN-backed static + SSR hosting
Build	Turbopack (https://turbo.build/pack)	Blazing-fast development builds
Linting	ESLint 9 (https://eslint.org/)	Code quality enforcement

Getting Started

Prerequisites

- Node.js ≥ 18
- npm ≥ 9
- A **Firebase** project with Authentication & Firestore enabled
- A **Google Maps** API key (Places, Maps JavaScript, Geocoding)
- A **Google Gemini** API key

Installation

```
# Clone the repository
git clone <your-repo-url>
cd movemate-pro

# Install dependencies
npm install
```

Environment Setup

Create a `.env.local` file in the project root:

```
NEXT_PUBLIC_FIREBASE_API_KEY=your_firebase_api_key
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=your_project.firebaseio.com
NEXT_PUBLIC_FIREBASE_PROJECT_ID=your_project_id
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=your_project.firebaseiostorage.app
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID=your_sender_id
NEXT_PUBLIC_FIREBASE_APP_ID=your_app_id
NEXT_PUBLIC_GOOGLE_MAPS_KEY=your_google_maps_key
GEMINI_API_KEY=your_gemini_api_key
```

Run Development Server

```
npm run dev
```

Open <http://localhost:3000> (<http://localhost:3000>) in your browser.

Production Build

```
npm run build
npm start
```

🗺 Future Features Roadmap

Phase 1 — Core Enhancements

Feature	Description	Potential Tech
Push Notifications	Real-time booking status updates	Firebase Cloud Messaging (FCM)
User Data Persistence	Secure storage for user preferences and detailed activity logs	Cloud Firestore / Firebase Auth
Payment Integration	Online payments at checkout	Stripe SDK / PayPal API
Email Confirmations	Booking receipts & reminders	SendGrid / Firebase Extensions
Multi-Language Support	i18n for Finnish, English, etc.	next-intl / react-i18next
PWA Install Prompt	Native app-like install experience	next-pwa, Service Workers

Phase 2 — Advanced Capabilities

Feature	Description	Potential Tech
Real GPS Tracking	Live driver location on map	Socket.IO / Firebase Realtime DB
Driver Portal	Separate dashboard for drivers	Next.js role-based routing
Photo Evidence	Before/after photos of items	Firebase Storage + Cloud Vision
Chat Support	In-app messaging with driver/support	Firebase Realtime DB / Stream Chat
Ratings & Reviews	Post-move rating system	Firestore + aggregation functions

Phase 3 — Scale & Intelligence

Feature	Description	Potential Tech
Dynamic Pricing	Surge pricing based on demand/season	Cloud Functions + ML models
Route Optimization	Multi-stop route planning	Google Routes API / OR-Tools
Inventory History	Save & reuse past inventories	Firestore subcollections
Admin Dashboard	Full back-office management panel	React Admin / Retool
Analytics & Insights	User behavior, conversion funnels	Google Analytics 4 / Mixpanel
Video Scanning	Walkthrough video inventory analysis	Gemini 1.5 Pro (video input)

Phase 4 — Enterprise & Platform

Feature	Description	Potential Tech
Multi-Tenant	White-label for moving companies	Firebase multi-project architecture
Fleet Management	Vehicle tracking, scheduling, dispatch	Google Fleet Engine
Automated Quotes	Instant AI-generated quotes via WhatsApp	Twilio API + Gemini
Insurance Claims	Damage reporting and claims workflow	Cloud Functions + Firestore
API Gateway	Public API for partners	Cloud Endpoints / Apigee

📄 Available Scripts

Command	Description
npm run dev	Start dev server on localhost:3000
npm run build	Create production build
npm start	Serve production build
npm run lint	Run ESLint checks

📁 Firebase Setup

Firestore Security Rules

The project includes pre-configured security rules (`firestore.rules`) that restrict data access to authenticated users.

Firestore Indexes

Composite indexes are defined in `firestore.indexes.json` for optimized query performance on bookings and addresses.

Deploy to Firebase

```
npmx firebase deploy
```

🤝 Contributing

- 1. Fork the repository
- 2. Create a feature branch (git checkout -b feature/amazing-feature)
- 3. Commit your changes (git commit -m 'Add amazing feature')
- 4. Push to the branch (git push origin feature/amazing-feature)
- 5. Open a Pull Request

 License

This project is private and proprietary. All rights reserved.

<p align="center">
Built with ❤ using Next.js, Firebase & Google AI
</p>